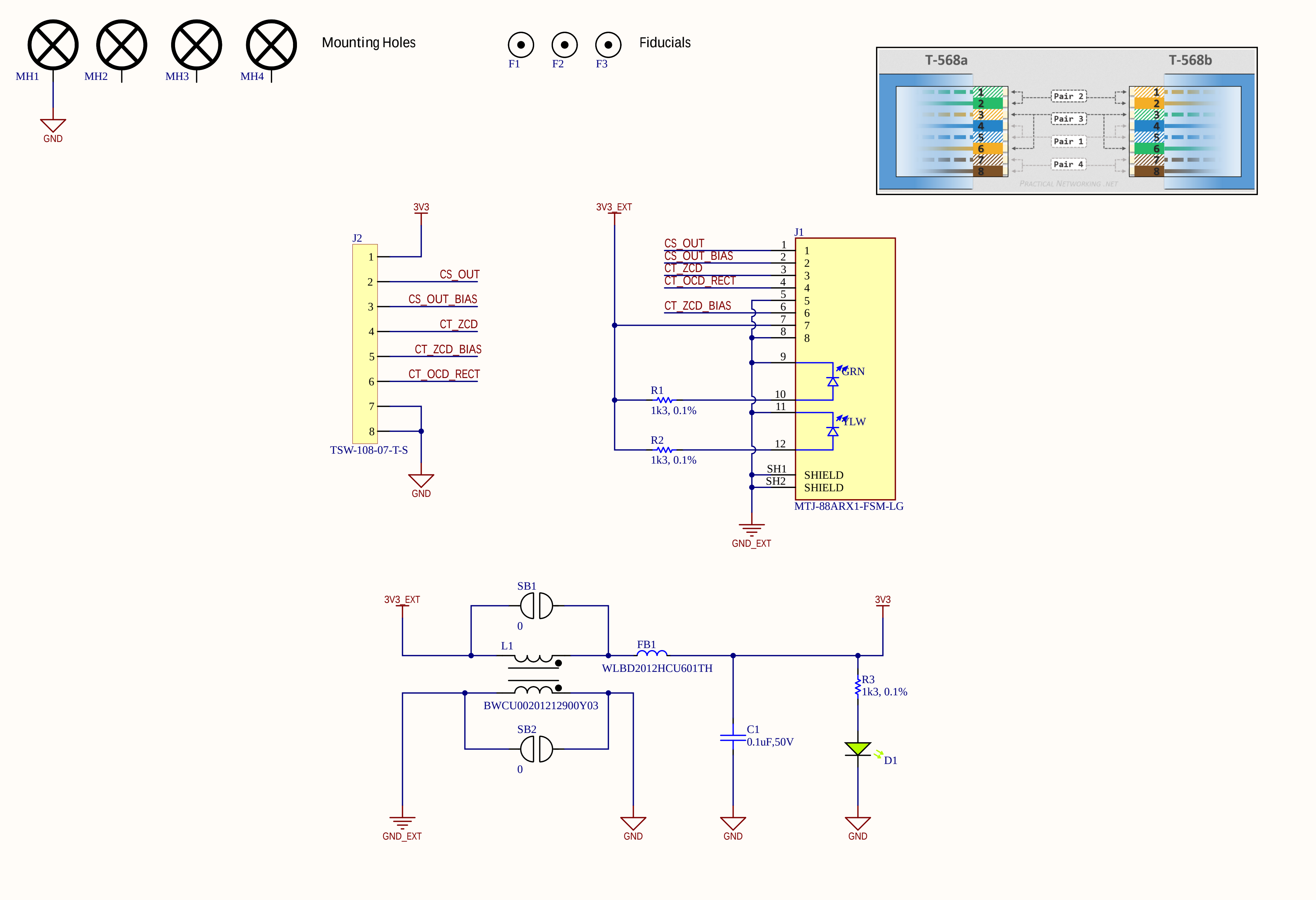
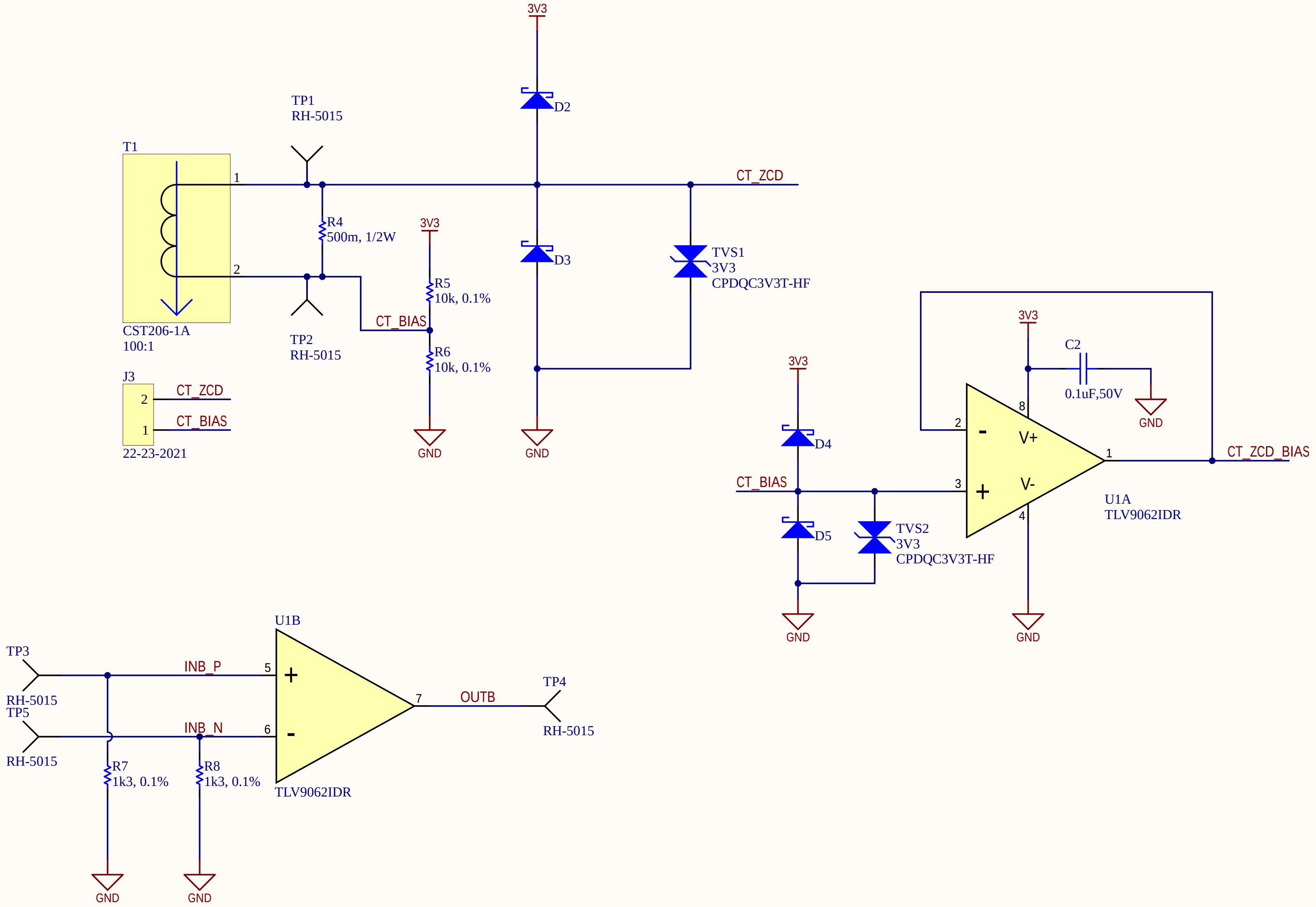


| | | |
|---|---|---|
| Title: Emma Mk1 DRSSTC CS | github.com/muJoseph | |
| Description: External Interface | | |
| File: INTERFACE.SchDoc | | Rev: Bravo |
| Size:Letter | Date: 7/27/2026 | EE: Joe Corteo | Sheet 1 of 4 |

- Set burden resistor, R4 <= VCC/2 * (Ns/Np) *(1/Ip(pk))

Where:
- Ns: CT secondary turns
- Np: CT primary turns
- VCC: Supply voltage



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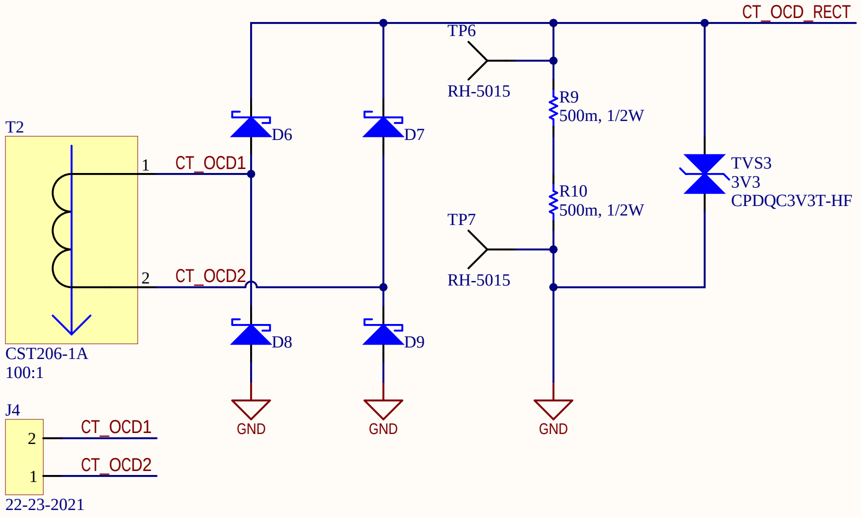
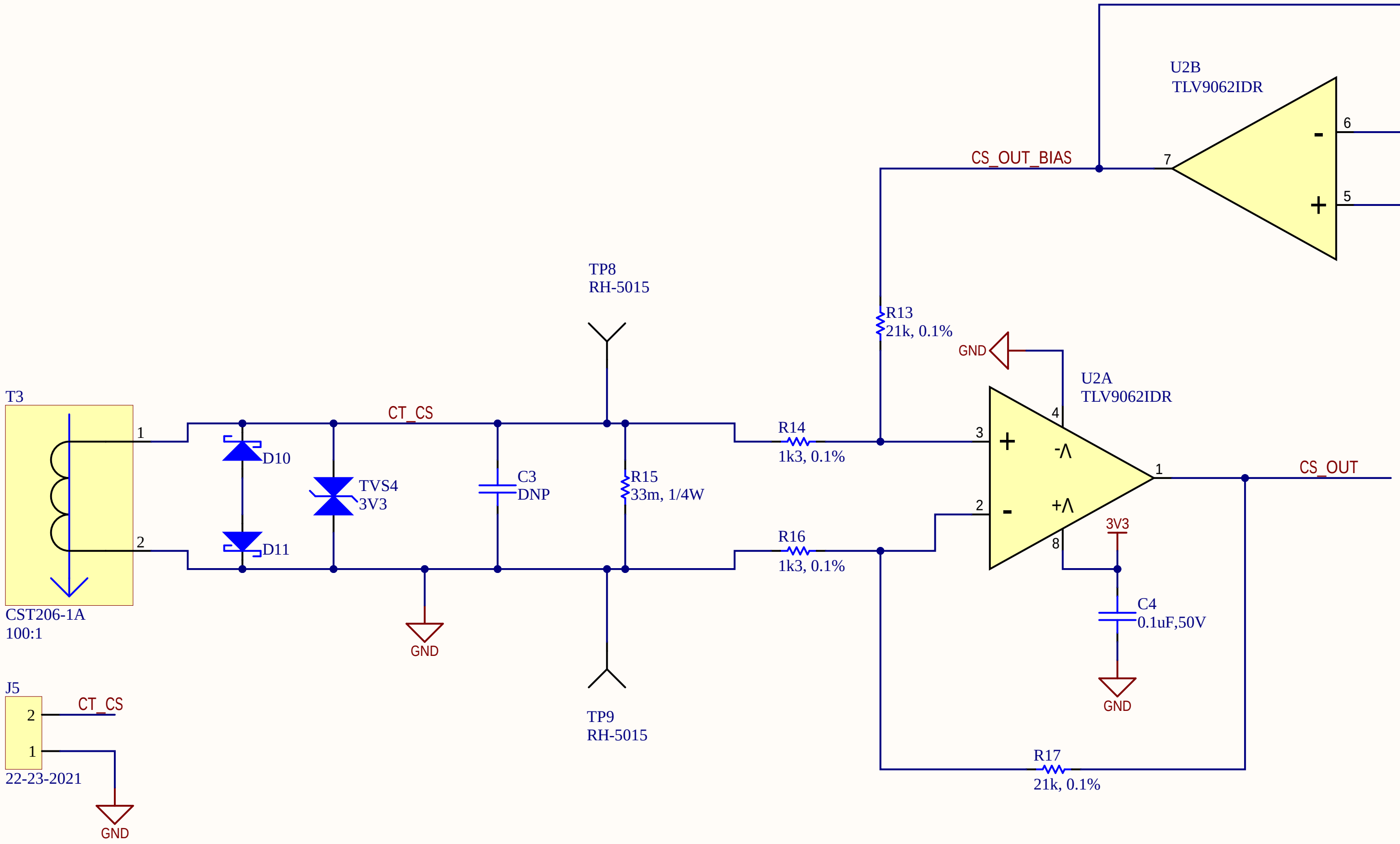
|---|---|---|
| Title: Emma Mk1 DRSSTC CS | github.com/muJoseph | |
| Description: Zero Crossing Detection | | |
| File: ZCD.SchDoc | | Rev: Bravo |
| Size:Letter | Date: 7/27/2026 | EE: Joe Corteo |
| | | Sheet 2 of 4 |



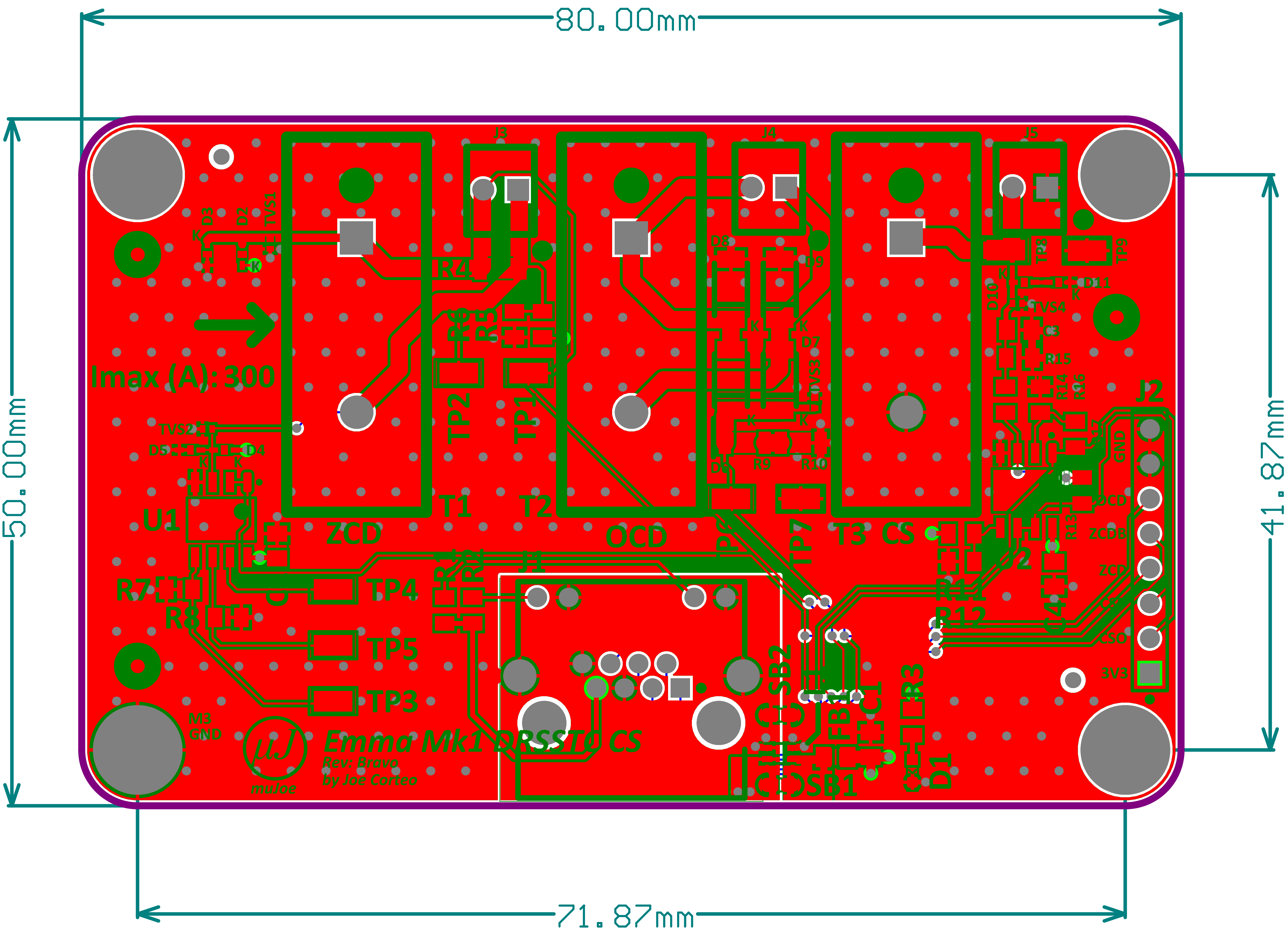
Title: Emma Mk1 DRSSTC CS
github.com/muJoseph
Description: Overcurrent Detection
File: OCD.SchDoc
Rev: Bravo
Size:Letter
Date: 7/27/2026
EE: Joe Corteo
Sheet 3 of 4

- Select burden resitor to achieve max shunt voltage of 100 mV at peak primary current: R15 <= 100mV/(Ipri(pk)/Ns)

Where:
Ns: CT secondary windings



| | | |
|---|---|---|
| Title: Emma Mk1 DRSSTC CS | github.com/muJoseph | |
| Description: Low-side bidirectional current sense | | |
| File: SENSE.SchDoc | | Rev: Bravo |
| Size:Letter | Date: 7/27/2026 | EE: Joe Corteo | Sheet 4 of 4 |


80.00mm
50.00mm
41.87mm
71.87mm
Imax (A): 300
ZCD
OCD
T1
T2
T3
CS
J2
J3
J4
J5
U1
U2
TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TP9
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
R15
R16
C1
C2
C3
C4
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
TVS1
TVS2
TVS3
TVS4
FB1
SB1
SB2
GND
OCD
ZCDB
ZCD
CS
CSO
3V3
M3
GND
μJ
muJoe
Emma Mk1 DRSSTC CS
Rev: Bravo
by Joe Corteo